Course Name:	Web Programming Laboratory (216U01L405)	**Semester:**	**IV**
Date of Performance:	**29/ 01/ 2026**	**DIV/ Batch No:**	**B – 1**
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Experiment No: 1

Title: Design and Development of a Web Portal using HTML5

Aim of the Experiment: To design and develop a multi-page, case-study-based web portal using HTML5, demonstrating proper document structure, semantic elements, formatting, lists, links, images, tables.

Objectives of the Experiment:

1. Develop structured and semantic HTML5 documents
2. Create interactive and media-rich web pages
3. Apply real-world problem context to web development
4. Build a basic but complete front-end web portal

COs to be achieved:

CO1: Design dynamic web pages using various HTML tags.

Books/ Journals/ Websites references:

1. Students should write

Theory:

Students should write about HTML 5 (citation needed for content taken (IEEE or APA7 style))

Problem statement: Urban citizens often face difficulties in accessing accurate and up-to-date public transport information such as route details, stop sequences, service schedules, and delay notifications. Existing systems are either fragmented across multiple platforms or lack a user-friendly interface for quick access

Task:

Task 1: Transport Portal Home Structure
Use Case:
Design the **homepage layout** of the public transport information system

Requirements:
- Page title reflecting the transport portal
- Header with application name (e.g., City Transport Info System)
- Navigation menu: Home, Routes, Schedules, Stops, Feedback
- Main section describing portal objectives
- Footer with transport authority notice

Deliverable: index.html
Skills Tested: HTML document structure, semantic tags

Task 2: Route Information Card
Use Case:
Display **basic route details** in a card-style format.
Requirements:
- **Route Number**
- **Source and Destination**
- **Total Stops**
- **Route status (Active / Diverted)**
- **Important route highlighted**

Deliverable: route-card.html
Skills Tested: Formatting & text-level tags, block-level layout

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Task 3: Route Stops List
Use Case: Show all stops covered by a selected route.
Requirements:
- Ordered list of stops
- Description list showing stop name and landmark
- Clear hierarchy of route → stops

Deliverable: route-stops.html
Skills Tested: List tags, structured content presentation

Task 4:
Transport Type & Category Info
Use Case:
Display different **transport service types**.

Requirements:
- Categories like Regular, Express, AC, Night Service
- Highlight premium or emergency services
- Use small text for service notes

Deliverable: service-category.html
Skills Tested: Text formatting, inline elements

Task 5: Citizen Feedback Form
Use Case: Collect citizen feedback or complaints regarding transport services.
Requirements:
- Input fields for name and contact
- Dropdown for route selection
- Feedback textarea
- Mandatory fields using `required`
- Submit button

Deliverable: feedback-form.html
Skills Tested: Form tags, global attributes, user input design

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Task 6: Daily Transport Schedule Table
Use Case:
Display **route-wise transport schedules**.
Requirements:
- **Table caption describing schedule**
- **Columns: Route No, Stop Name, Arrival Time, Departure Time**
- **At least 3 sample data rows**

Deliverable: schedule-table.html
Skills Tested: Table tags, tabular data organization

Task 7: Route Map & Stop Images

Use Case:

Provide a **visual representation of routes and major stops**.

Requirements:

- Display route map image
- Use image map to click on major stops
- Proper `alt` text for accessibility

Deliverables: route-map.html

Skills Tested:

Image handling, image maps

Task 8: Useful Transport Links

Use Case:

Provide **important transport-related links**.

Requirements:

- Links to city transport authority
- Emergency helpline link
- Timetable PDF link
- External links open in a new tab

Deliverable: transport-links.html

Skills Tested: Hyperlink tags, navigation logic

Code:

Task 1: Railway Transport Information System Portal - Home Structure

index.html

index.html

Experiment1 > index.html > html > body > footer > center > small > br

```html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <title>Railway Transport Information System</title>
</head>
<body>

<header>
    <center>
        <h1>🚆 Railway Transport Information System</h1>
        <p><em><b>Live railway routes, stations, schedules & updates</b></em></p>
    </center>
</header>

<hr>

<nav>
    <center>
        <strong>
            <a href="index.html">🏠 Home</a> |
            <a href="route-card.html">🗺️ Routes</a> |
            <a href="schedule-table.html">⏱️ Schedule</a> |
            <a href="route-stops.html">🚉 Stations</a> |
            <a href="feedback-form.html">📝 Feedback</a>
        </strong>
    </center>
```

```html
</nav>

<hr>

<main>
    <section>
        <center>
            <figure>
                <img src="Mumbai Railway Banner Image.jpg" alt="Indian Railway Network" width="1500">
                <figcaption>
                    <small>Indian Railway Route Information Platform</small>
                </figcaption>
            </figure>
        </center>

        <blockquote><center>
        This platform provides passengers with
        <strong>real-time railway information</strong> including
        <mark>routes</mark>, <mark>station details</mark>,
        <mark>schedules</mark>, and <mark>service updates</mark>,
        similar to modern railway information systems.
        </blockquote></center>
    </section>
</main>
```

```html
</main>

<hr>

<footer>
    <center>
        <small>
            © Railway Transport Authority <br>
            For passenger information only
        </small>
    </center>
</footer>

</body>
</html>
```

Task 2: Route Information Card
route-card.html

Experiment1 > route-card.html > html > body > center > fieldset > table > tr > td

```html
<!DOCTYPE html>
<html>
<head>
    <title>Route Information Card</title>
</head>
<body>

<center>
    <h1>🚆 Route Information Card</h1>
</center>

<hr>

<center>
    <fieldset>
    <legend><strong>Route Details</strong></legend>

    <table cellpadding="8">
        <tr>
            <td><strong>Route No</strong></td>
            <td>:</td>
            <td><mark>101</mark></td>
        </tr>
        <tr>
            <td><strong>Source</strong></td>
            <td>:</td>
            <td>Andheri</td>
```

```html
            </tr>
            <tr>
                <td><strong>Destination</strong></td>
                <td>:</td>
                <td>Churchgate</td>
            </tr>
            <tr>
                <td><strong>Total Stops</strong></td>
                <td>:</td>
                <td>18</td>
            </tr>
            <tr>
                <td><strong>Status</strong></td>
                <td>:</td>
                <td><mark>✅ Active</mark></td>
            </tr>
        </table>

        <br>

        <center>
            <figure>
```

```html
        <center>
            <figure>
                <img src="Railway Route Image.jpg" alt="Railway Route Overview" width="800">
                <figcaption>
                    <small>Railway route overview for Route 101</small>
                </figcaption>
            </figure>
        </center>

    </fieldset>

    </center>

    <hr>

    </body>
    </html>
```

Task 3: Route Stops List
route-stops.html

Experiment1 > route-stops.html > html > body > center > fieldset

```html
<!DOCTYPE html>
<html>
<head>
    <title>Route 101 - Stops</title>
</head>
<body>

<center>
    <h1>📍 Route 101 – Stops Information</h1>
</center>

<hr>

<center>
    <fieldset>
        <legend><strong>🚉 Stop Sequence</strong></legend>

        <ol>
            <li><b>Andheri Station</b></li>
            <li><b>Bandra</b></li>
            <li><b>Dadar</b></li>
            <li><b>Mumbai Central</b></li>
            <li><b>Churchgate</b></li>
        </ol>
    </fieldset>
</center>

```

```html
<br>

<center>
    <fieldset>
        <legend><strong>📌 Stop Landmarks</strong></legend>

        <dl>
            <dt><strong><b>Andheri</b></strong></dt>
            <dd><center>Near Metro Station</center></dd>

            <dt><strong><b>Bandra</b></strong></dt>
            <dd><center>Near Linking Road</center></dd>

            <dt><strong><b>Dadar</b></strong></dt>
            <dd>Central Railway Hub</dd>
        </dl>
    </fieldset>
</center>

<hr>

<center>
    <small>Landmark information for passenger convenience</small>
</center>

<center>
    <figure>
        <img src="Mumbai Railway Route Stops and Landmarks.jpg" alt="Railway Route Overview" width="800">
        <figcaption>
            <small><b> Major railway stations along Route 101 </b></small>
        </figcaption>
    </figure>
</center>

</body>
</html>
```

Task 4: Transport Type & Category Info
service-category.html

Experiment1 > service-category.html > html > body > center > fieldset

```html
<!DOCTYPE html>
<html>
<head>
    <title>Transport Service Types</title>
</head>
<body>

<center>
    <h1>🚌 Transport Service Types</h1>
    <p><em>Available public transport categories</em></p>
</center>

<hr>

<center>
    <fieldset>
        <legend><strong>Service Categories</strong></legend>

        <ul>
            <li><b>Regular Service</b></li>
            <li><b>Express Service</b></li>
            <li><mark>❄️ <b>AC Premium Service</b></mark></li>
            <li><b>🌙 Night Service</b> <small>(Limited stops)</small></li>
        </ul>

    </fieldset>
</center>
```

```html

<hr>

<center>
    <small>Service availability may vary based on route and time</small>
</center>

<center>
    <figure>
        <img src="Types of Trains.jpg" alt="Different Railway Service Types" width="800">
        <figcaption>
            <small>Different categories of passenger railway services</small>
        </figcaption>
    </figure>
</center>

<br>

</body>
</html>

```

Task 5: Citizen Feedback Form
feedback-form.html

Experiment1 > feedback-form.html > html > body > center

```html
<!DOCTYPE html>
<html>
<head>
    <title>Citizen Feedback Form</title>
</head>
<body>

<center>
    <h1> Citizen Feedback Form</h1>
    <p><em>Your feedback helps us improve transport services</em></p>
</center>

<hr>

<center>
<form>
    <fieldset>
        <legend><strong> Feedback Details</strong></legend>

        <table cellpadding="8">
            <tr>
                <td><strong>Name</strong></td>
                <td>:</td>
                <td>
                    <input type="text" required>
                    <mark>*</mark>
                </td>
```

```
            </tr>

            <tr>
                <td><strong>Contact</strong></td>
                <td>:</td>
                <td>
                    <input type="text" required>
                    <mark>*</mark>
                </td>
            </tr>

            <tr>
                <td><strong>Route</strong></td>
                <td>:</td>
                <td>
                    <select>
                        <option>Route 101</option>
                        <option>Route 202</option>
                        <option>Route 303</option>
                    </select>
                </td>
            </tr>
```

```html

            <tr>
                <td><strong>Feedback</strong></td>
                <td>:</td>
                <td>
                    <textarea rows="4" cols="25" required></textarea>
                    <mark>*</mark>
                </td>
            </tr>

            <tr>
                <td colspan="3" align="center">
                    <input type="submit" value="Submit Feedback">
                </td>
            </tr>
        </table>

        <center>
            <small><mark>*</mark> Mandatory fields</small>
        </center>

    </fieldset>
```

```html
    </fieldset>
</form>
</center>

<center>
    <figure>
        <img src="Citizen Feedback Form.jpg" alt="Railway Passenger Feedback and Assistance" width="500">
        <figcaption>
            <small>Passenger feedback and assistance services</small>
        </figcaption>
    </figure>
</center>

<br>

<hr>
```

Task 6: Daily Transport Schedule Table

schedule-table.html

Experiment1 > schedule-table.html > html > body > center > fieldset > table > tr > td

```html
<!DOCTYPE html>
<html>
<head>
    <title>Daily Transport Schedule</title>
</head>
<body>

<center>
    <h1>⏰ Daily Transport Schedule</h1>
    <p><em>Route-wise arrival and departure timings</em></p>
</center>

<hr>

<center>
    <fieldset>
        <legend><strong>🚆 Route 101 Schedule</strong></legend>

        <table border="1" cellpadding="8">
            <caption><strong>Daily Transport Schedule</strong></caption>

            <tr>
                <th>Route No</th>
                <th>Stop Name</th>
                <th>Arrival Time</th>
                <th>Departure Time</th>
            </tr>
```

```html

        <tr>
            <td>101</td>
            <td>Andheri</td>
            <td>08:00</td>
            <td>08:05</td>
        </tr>

        <tr>
            <td>101</td>
            <td>Bandra</td>
            <td>08:30</td>
            <td>08:35</td>
        </tr>

        <tr>
            <td>101</td>
            <td>Churchgate</td>
            <td>09:15</td>
            <td>–</td>
        </tr>
    </table>
```

```html

    </fieldset>
</center>

<hr>

<center>
    <small>Schedule timings are subject to traffic conditions</small>
</center>

<center>
    <figure>
        <img src="Mumbai Railway Route Stops.jpg" alt="Railway Route Overview" width="500">
        <figcaption>
            <small>Railway route Stops and Landmarks</small>
        </figcaption>
    </figure>
</center>

</body>
</html>
```

Task 7: Route Map & Stop Images

route-map.html

Experiment1 > route-map.html > html > body

```html
<!DOCTYPE html>
<html>
<head>
    <title>Route Map</title>
</head>
<body>

<center>
    <h1>🗺️ Route Map & Major Stops</h1>
    <p><em>Visual representation of transport routes and stations</em></p>
</center>

<hr>

<center>
    <fieldset>
        <legend><strong>📍 City Transport Route Map</strong></legend>

        <figure>
            <img src="https://erail.in/images/net-map/mumbai-local-trains.jpg" alt="City Transport Route Map" usemap="#stops" width="800">

            <figcaption>
                <small>
                    Click on highlighted areas to view major stops
                </small>
            </figcaption>
        </figure>

        <map name="stops">
            <area shape="rect" coords="50,50,150,150" alt="Andheri">
            <area shape="rect" coords="200,50,300,150" alt="Bandra">
        </map>

```

```
    </fieldset>
</center>

<hr>

<center>
    <small>
        Map is for reference purposes only <br>
        Source: City Transport Authority
    </small>
</center>

</body>
</html>
```

Task 8: Useful Transport Links
transport-links.html

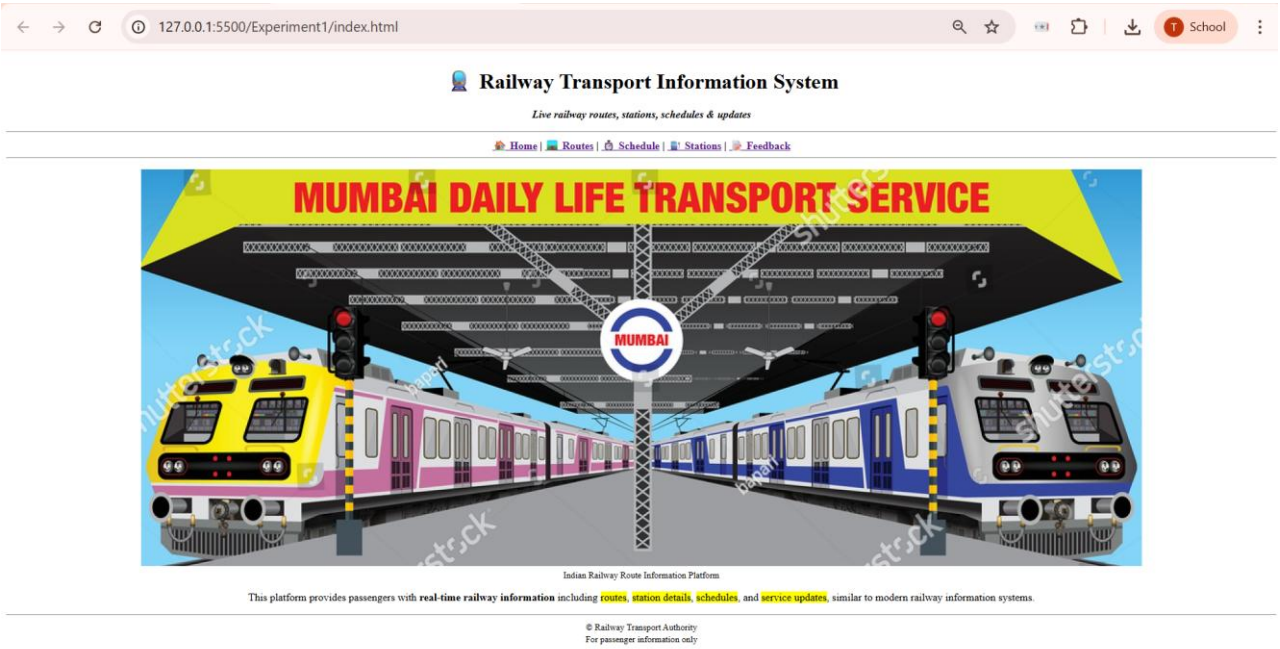
Experiment1 > transport-links.html > ...

```html
<!DOCTYPE html>
<html>
<head>
    <title>Important Transport Links</title>
</head>
<body>

<center>
    <h1>🔗 Important Transport Links</h1>
    <p><em>Quick access to essential transport resources</em></p>
</center>

<hr>

<center>
    <fieldset>
        <legend><strong>Useful Links</strong></legend>

        <ul>
    <li>
        🌐 <a href="https://shorturl.at/TifvA" target="_blank">
        City Transport Authority</a>
    </li>

    <li>
        🚨 <a href="https://url-shortener.me/AJNE" target="_blank">
        Emergency Helpline</a>
        <small>(24×7 support)</small>
    </li>

    <li>
        📄 <a href="Mumbai Railway Timetable.pdf" target="_blank">
        Download Timetable</a>
```

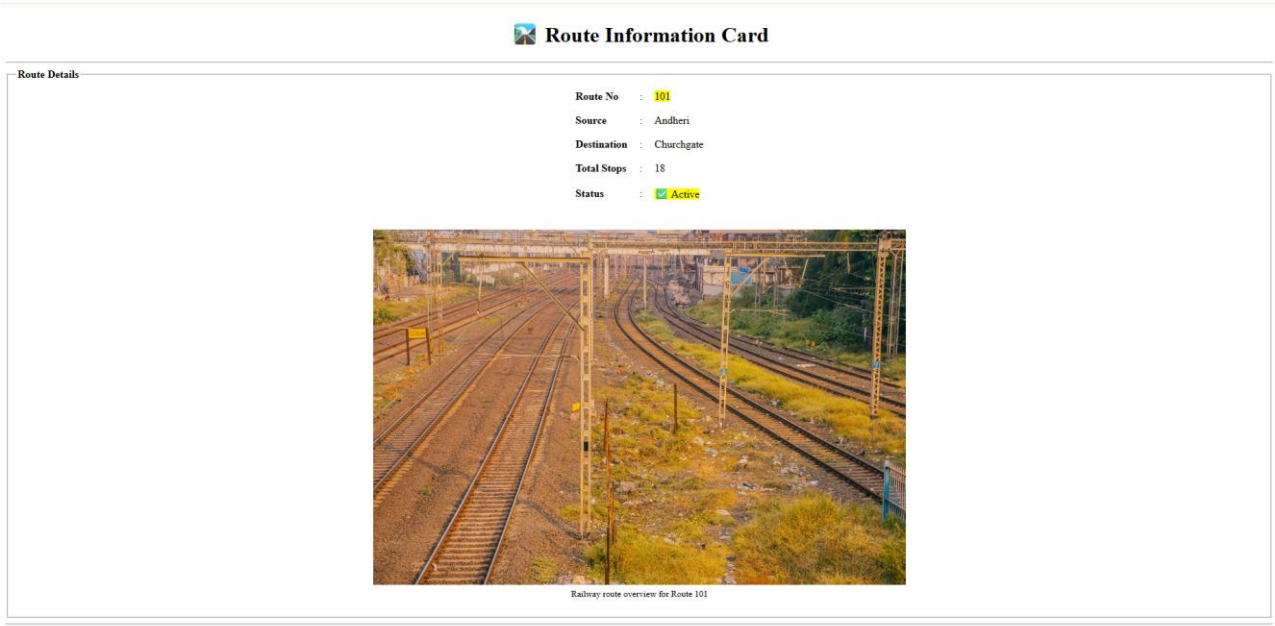
```html
        </li>
    </ul>

        </fieldset>
    </center>

    <hr>

    <center>
        <small>
            External links open in a new tab <br>
            Provided for public convenience
        </small>
    </center>

    <center>
        <figure>
            <img src="Transport Links.jpg" alt="Railway Information Resources" width="600">
            <figcaption>
                <small>Official railway information and passenger support resources</small>
            </figcaption>
        </figure>
    </center>

    <hr>


</body>
</html>
```

Output / Screenshot:

Task 1: Railway Transport Information System Portal - Home Structure



Task 2: Route Information Card



Task 3: Route Stops List

Route 101 – Stops Information

Stop Sequence

1. Andheri Station
2. Bandra
3. Dadar
4. Mumbai Central
5. Churchgate

Stop Landmarks

Andheri
Near Metro Station
Bandra
Near Linking Road
Dadar
Central Railway Hub

Landmark information for passenger convenience

Major railway stations along Route 101

Task 4: Transport Type & Category Info

Transport Service Types

Available public transport categories

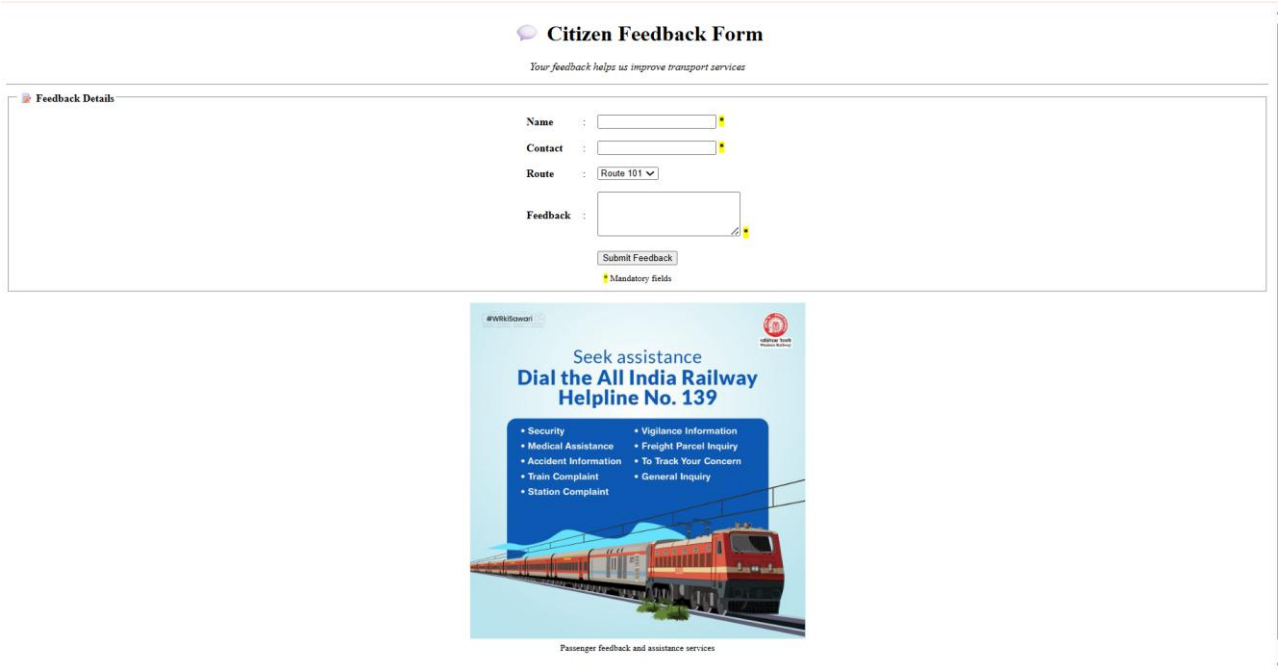
Service Categories

- Regular Service
- Express Service
- AC Premium Service
- Night Service (Limited stops)

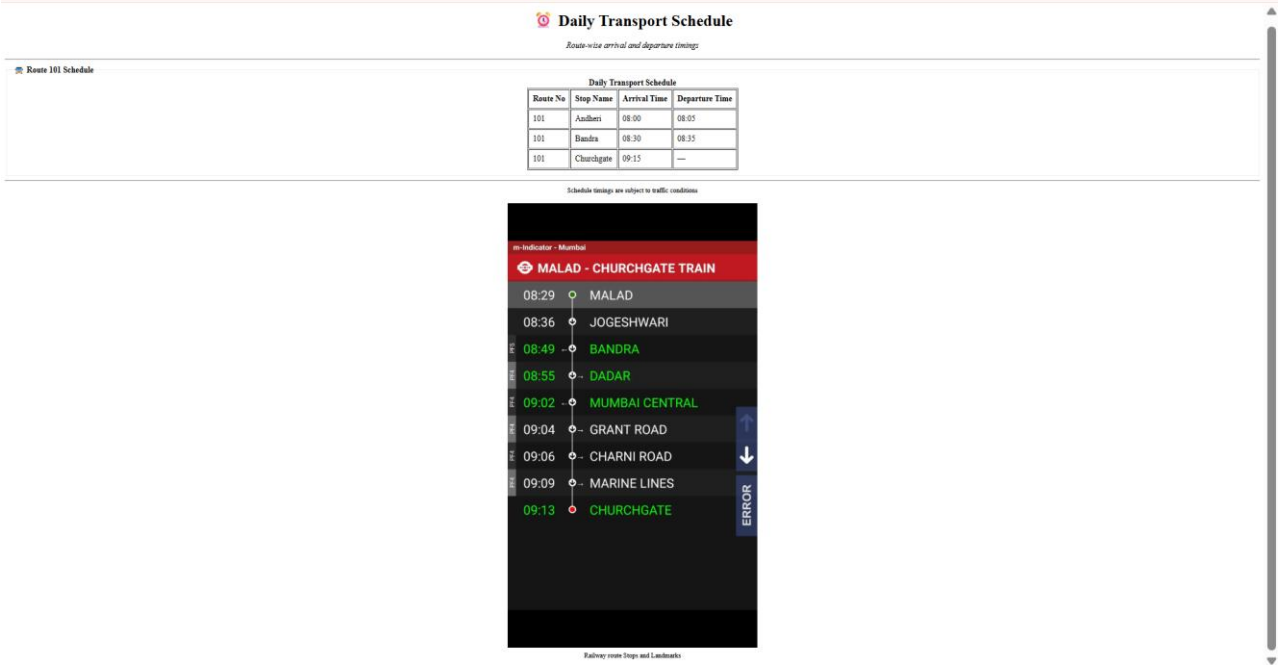
Service availability may vary based on route and time

Different categories of passenger railway services

Task 5: Citizen Feedback Form



Task 6: Daily Transport Schedule Table



Task 7: Route Map & Stop Images

Route Map & Major Stops

Visual representation of transport routes and stations

City Transport Route Map

Click on highlighted areas to view major stops

Map is for reference purposes only
Source: City Transport Authority

Task 8: Useful Transport Links

Important Transport Links

Quick access to essential transport resources

Useful Links

- City Transport Authority
- Emergency Helpline (24×7 support)
- Download Timetable

External links open in a new tab
Provided for public convenience

Official railway information and passenger support resources

Post Lab Subjective/Objective type Questions:

1. Write HTML code to create a hyperlink that opens in a new tab.

Ans.

To open a hyperlink in a new browser tab, the target="_blank" attribute is used inside the <a> tag.

Code:

```
<a href="https://www.indianrailways.gov.in" target="_blank">
   Visit Indian Railways Website
</a>
```

2. Differentiate between block-level and inline elements with examples.

Ans.

Block-Level Elements	Inline Elements
Block-level elements occupy the **entire width** of the page.	Inline elements occupy **only the space required** by their content.
They always start on a **new line**.	They do **not** start on a new line.
They are mainly used to define the **structure and layout** of a web page.	They are used to format or style **part of the content** within a line.
Height and width can be applied.	Height and width cannot be applied directly.
Examples: `<div>`, `<p>`, `<h1>`, `<ul>`, `<table>`	Examples: `<span>`, `<a>`, `<strong>`, `<em>`, `<img>`

Code:

```
<p>This is a block-level element.</p>
<a href="#">This is an inline element.</a>
```

3. Why are semantic tags preferred over tags?

Ans.

Semantic tags are preferred over tags because they clearly describe the meaning and purpose of the content they contain, whereas is a generic container with no semantic meaning.

Reasons:

- Semantic tags improve code readability and structure.
- They help search engines understand page content better (SEO).
- They enhance accessibility for screen readers.
- They make maintenance easier for developers.

Examples of Semantic Tags:

- <header> – Defines the header section
- <nav> – Defines navigation links
- <section> – Groups related content
- <article> – Independent content

- <footer> – Defines footer information

Comparison Example:

<header>This is a header</header>
This is a header

4. Why is <iframe> preferred over <frame> in modern HTML?

Ans.

<iframe> is preferred over <frame> because <frame> and <frameset> are deprecated in HTML5, while <iframe> is fully supported and flexible.

Reasons:

- <iframe> is supported in HTML5, whereas <frame> is obsolete.
- <iframe> can be used inside the normal HTML document flow.
- <iframe> allows embedding external content (webpages, videos, maps) easily.
- <frame> caused navigation, accessibility, and SEO issues.

Example:

<iframe src="https://www.indianrailways.gov.in" width="600" height="400"></iframe>

5. Evaluate the importance of using alt attribute in images.

Ans.

The alt attribute provides alternative text for an image when the image cannot be displayed and is essential for accessibility and usability.

Importance of alt attribute:

- Helps screen readers describe images to visually impaired users.
- Displays descriptive text if the image fails to load.
- Improves search engine optimization (SEO) by describing image content.
- Makes web pages more accessible and user-friendly.

Example:

<img src="train.png" alt="Indian railway passenger train">

6. Suggest two additional features that can improve the given web portal and explain how HTML5 supports them.

Ans.

Feature 1: Live Announcement & Alerts Section

Description:
A section to display important railway announcements such as delays, platform changes, or service disruptions.

HTML5 Support:

- HTML5 semantic tags like <section> and <article> can be used to structure announcements.
- <mark> can highlight important alerts.
- <time> tag can display the date and time of announcements clearly.

Example:

```
<article>
   <mark>Train delayed by 15 minutes</mark>
   <time>10:30 AM</time>
</article>
```

Feature 2: Station-wise Multimedia Information

Description:
Providing images or videos of major railway stations to help passengers identify stations easily.

HTML5 Support:

- <figure> and <figcaption> can be used for structured image display.
- <audio> and <video> tags allow embedding multimedia without plugins.
- Improves user engagement and understanding.

Example:

```
<figure>
   <img src="station.png" alt="Railway station platform">
   <figcaption>Major railway station platform</figcaption>
</figure>
```

Conclusion and Discussion:

In this experiment, a **multi-page Railway Transport Information System** was successfully designed and developed using **HTML5**. The web portal provides essential railway-related information such as routes, stations, schedules, service categories, feedback forms, maps, and useful links. Proper use of **HTML5 semantic elements**, lists, tables, forms, images, and hyperlinks helped in creating a structured and user-friendly web interface.

The experiment improved understanding of **document structure, semantic tags, form elements, image handling, and tabular data presentation** in HTML. Special care was taken to enhance the visual appearance of the website using **only HTML elements**, without using CSS or JavaScript, as per the laboratory constraints.

Feedback from fellow classmates and faculty suggested that the portal is **well-organized, easy to navigate, and closely resembles real-world railway information systems**. The use of meaningful images and semantic tags made the portal more intuitive and accessible. Overall, this experiment helped in gaining practical experience in front-end web development and demonstrated how HTML5 alone can be effectively used to build a complete and functional web portal.